
UTOPI: Efficient Egocentric Long-Video Understanding in AR via User-Guided Token Pre-Compression

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Abstract

1 As long-video understanding continues to gain traction, Augmented Reality (AR)
2 devices emerge as a compelling deployment platform. However, egocentric videos
3 exhibit significant temporal redundancy, imposing substantial memory and compu-
4 tational burdens on resource constrained AR hardware. To address this challenge,
5 we introduce *UTOPI*, a plug-and-play pre-compression module designed specifi-
6 cally for egocentric long videos on AR platforms. First, we leverage user motion
7 signals to estimate inter-frame overlap and propose a viewpoint-shift-invariant
8 pruning algorithm. Second, we exploit eye-tracking sensors to identify user fix-
9 ations, enabling the separation of attended foreground content from background
10 and its preferential retention via a fine-tuned neural network-based adjustment
11 mechanism. Across multiple benchmarks and model backbones, *UTOPI* achieves
12 up to 95% compression while preserving performance.

13 1 Introduction

14 Augmented Reality (AR) is emerging not only as a new user interface but also as a powerful machine
15 learning (ML) platform that integrates sensing, perception, memory, and speech. By tightly coupling
16 the digital and physical worlds, AR enables real-time context extraction and decision making across
17 domains such as healthcare [23, 16, 43], education [62, 6, 59], and industry [17]. Beyond immersive
18 interaction, AR devices continuously generate rich multimodal data. With cameras, eye tracking, and
19 motion sensors, they capture egocentric video together with pose and behavioral signals, creating new
20 opportunities for long-context ML models to reason over user attention and environmental context.

21 As shown in Figure 1 (a), consider an AR head-mounted display (HMD) used as a memory assistant
22 in daily activities. The device continuously captures and encodes multimodal inputs with ML models,
23 and a brief user glance at a drawer must be reliably bound to the corresponding visual features. Later,
24 answering a query such as “How many plates in my drawer?” requires an episodic memory module
25 that can index temporal embeddings, retrieve the correct event, and interface with a downstream
26 processing engine (e.g., a large model). This capability of *long video understanding* (LVU) transforms
27 AR from a passive sensing platform into an intelligent personal assistant with long-context reasoning,
28 marking a critical step toward next-generation AR applications. In education, students can revisit past
29 demonstrations to reinforce learning; in industry, recorded tasks can be replayed to identify errors;
30 and in daily life, AR can serve as a memory aid for locating items such as keys.

31 However, LVU places severe demands on resource-limited AR hardware. For example, encoding a
32 100-second high-definition video sampled at 1 FPS with a typical vision-language model (VLM)
33 can require tens of gigabytes of memory, far exceeding the 8–16 GB capacity of current wearable
34 devices [46, 7]. Reducing the sampling rate lowers memory usage but often degrades representation

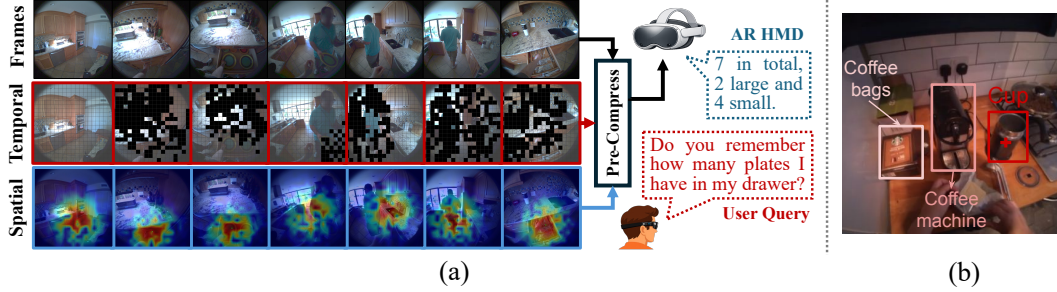


Figure 1: (a) UTOPI compresses temporal redundancy from viewpoint changes and spatial redundancy in user unattended regions. (b) An example on how user attention varies with gaze.

quality and downstream accuracy. Moreover, continuously running large encoder backbones for online encoding, retrieval, and reasoning is computationally prohibitive on AR devices. These constraints motivate new algorithms for efficient LVU in AR environments.

To address this challenge, we propose a *perception-guided approach* for efficient LVU in AR, inspired by human attention. Human vision is nonuniform, with highest fidelity near the focus of attention and lower sensitivity in the periphery [22, 56, 9]. In egocentric AR video, visual content is often tied to the user’s task, and prior studies show that user intent is reflected in the questions they ask and the answers they seek [66, 11, 26]. Thus, unattended regions often contain spatial redundancy and contribute less to downstream reasoning. AR streams also exhibit temporal redundancy, since smooth head and body motion creates substantial overlap across consecutive frames (Figure 1(a)). Our key idea is to exploit these spatial and temporal regularities by allocating computation and memory to salient regions and events while preserving information needed for LVU.

Building on this idea, we introduce *UTOPI*, a plug-and-play efficient egocentric Long-Video Understanding in AR via user-guided **T**oken **P**re-compression module. UTOPI reduces spatial redundancy by using gaze-informed attention and a lightweight saliency predictor to retain important tokens in attended regions. To reduce temporal redundancy, it preserves full-fidelity representations of overlapping regions at their first appearance, estimated from AR motion signals, and replaces redundant tokens in later frames with compressed surrogate tokens. We also introduce *EgoEye*, an AR-oriented LVU benchmark that reflects the **human-centric** and **attention-guided** nature of AR usage by aligning questions with users’ gaze patterns. Our contributions are summarized as follows:

- We introduce UTOPI, a plug-and-play user behavior-guided token pre-compression module that uses AR-native signals to compress visual tokens before the query stage, jointly reducing spatial and temporal redundancy to lower computation and memory costs.
- By integrating UTOPI with multiple VLMs, we enable video understanding under tight AR resource constraints. We also introduce EgoEye, an egocentric video benchmark with diverse sensor signals and both attention-driven and detail-oriented QA pairs.
- Across multiple egocentric LVU benchmarks and VLM backbones, UTOPI achieves consistent efficiency-accuracy gains, maintaining baseline accuracy while using as little as 5% of visual tokens on EgoSchema and substantially reducing on-device memory cost.

2 Background and Related Work

Efficient Long Video Understanding Contemporary vision-language models (VLMs), including proprietary systems [60, 49] and open-source models [70, 19, 10], encode images and videos into visual tokens and process them jointly with text. This unified framework supports perception, reasoning, instruction following, and dialogue, making VLMs common backbones for video understanding [41, 71, 37, 31, 58]. Recent works have also explored gaze-guided video understanding [53, 57], leveraging gaze signals to improve efficiency and temporal reasoning. However, VLMs are large and memory-intensive, especially because the decoding KV cache scales with the number of visual tokens. For AR assistants, user queries often require long temporal context, further increasing memory footprint and inference latency, which makes on-device deployment difficult.

Existing efficient video methods either sample temporal frames, such as VidF4 [36] and GenS [65], or select query-relevant spatial regions conditioned on the user question, such as DynTok [72]

and Q Frame [73]. Because these methods require the query, the video must be reprocessed for each question. In contrast, UTOPI performs query-agnostic pre-compression, allowing compressed visual tokens to be stored in the KV cache and reused across queries. Other query-free methods, such as DTD [64] and PruneVid [28], are not tailored to egocentric AR video and struggle with redundancy caused by continuous viewpoint changes. UTOPI instead uses user motion signals to enable viewpoint-shift-invariant token compression.

Prior work also shows that gaze is informative for modeling user behavior and intent [51, 48, 32, 34]. However, these methods mainly target accuracy and may introduce costly pipelines such as optical flow [34, 48]. Other gaze-based acceleration methods [55, 69, 14] often rely on simple zero-shot heuristics such as central cropping or foveated tokenization. In contrast, UTOPI uses gaze signals with a lightweight saliency gating predictor fine-tuned on EgoEye and EgoSchema, enabling more adaptive and accurate gaze-guided token compression.

Attention-Guided Question Formation in Egocentric AR

Human perception is not uniform across the visual field but instead prioritizes attended regions. Classic studies characterize visual attention as a “spotlight” [21, 54], often approximated by gaze position. Attention typically declines with increasing distance from this focus and can be modeled using functions such as a two-dimensional Gaussian [30], resulting in higher fidelity near the attended location and progressively lower fidelity in the periphery [20, 13]. This selectivity has direct implications for the types of questions an AR user is likely to ask. As illustrated in Figure 1 (b), a user wearing an AR HMD in a kitchen may fixate on a cup on the counter; objects that receive sustained attentional focus naturally become high-priority candidates for future queries, such as “Where did I leave the cup?”. In contrast, nearby objects with weaker attentional signals, shown with lighter bounding boxes in Figure 1 (b), are less likely to be referenced. From a modeling perspective, gaze thus provides a **useful prior for identifying semantically important tokens in egocentric visual streams**. Consistent with this view, prior work in cognitive psychology and vision science shows that gaze patterns are predictive of subsequent memory formation and information-seeking behavior [67, 33].

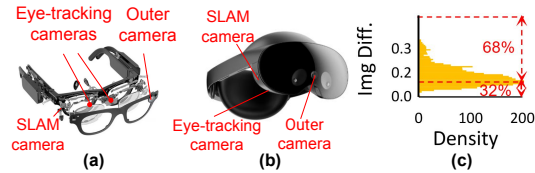


Figure 2: (a) Meta Aria glasses equipped with multiple cameras. (b) Meta Quest Pro headset, which functions as both an AR and VR device. (c) Distribution of image differences.

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Human Attention in AR Environment To study AR users’ attention patterns over time, we use the *ARIA Everyday* (AEA) dataset [39], collected with Meta Aria smart glasses. The dataset includes over 7.5 hours of egocentric multimodal recordings from more than 150 participants performing everyday activities, with synchronized first-person video, gaze trajectories, and head motion. As shown in Figure 2(c), we measure RGB pixel-wise differences between consecutive frames to quantify visual changes. About 32% of frame pairs show less than 9% change, indicating substantial temporal redundancy and strong correlation across adjacent frames. This suggests that much of the visual content in egocentric AR video remains stable over short intervals, enabling opportunities for efficient processing and computation reuse.

AR Hardware System Figure 2 shows a representative AR hardware configuration. Modern headsets typically include multiple front- and side-facing cameras for capturing the surrounding scene and user state, including gaze direction. Outward-facing cameras provide high-resolution scene views, e.g., 1408×1408 on Meta Aria Glass [2], while inward-facing cameras capture lower-resolution monochrome eye images. These sensors support gaze estimation [1, 47, 2] through analytic methods such as pupil–corneal reflection or learning-based approaches [38], providing key signals for user attention and interaction. Tracking cameras, together with onboard IMU measurements, further enable visual–inertial odometry and full 6-DoF head-pose estimation, similar to ORB-SLAM3 [12]. IMU and depth sensors are standard in modern AR systems such as HoloLens 2 [18], Apple Vision Pro [8], and Magic Leap 2 [42]. This sensing stack provides visual input, gaze information, and pose tracking, making AR platforms suitable for intention-aware perception and interaction.

However, mainstream AR headsets typically provide only 8–16 GB of on-device memory [46, 7], often shared by CPU and GPU under a unified memory architecture [46, 7, 42]. Since part of this

memory is reserved by the system and compute is limited by mobile-class chips, deploying LVU models on AR devices remains challenging and often leads to high multimodal response latency.

3 Methodology

3.1 UTOPI Overview

VLMs process a video by first sampling frames to form a frame set $\mathbf{I}$, and then encoding each frame using a ViT-based vision encoder to produce visual tokens $\mathbf{T} = \{\mathbf{T}^i\}_{i=1}^N$, where $\mathbf{T}^i = \{\mathbf{T}_{w,h}^i \mid w \in [1, W], h \in [1, H]\}$, i denotes the frame index, N is the number of sampled frames, and (w, h) represents the spatial location of token $\mathbf{T}_{w,h}^i$ within frame $\mathbf{I}^i$, with an $H \times W$ token grid. Positional embeddings $P(\cdot)$ are then applied to these tokens to obtain the embedded representation $\mathbf{T}_{\text{emb}} = P(\mathbf{T})$. Finally, $\mathbf{T}_{\text{emb}}$ is fed together with textual tokens into the language decoder to generate the final output. To reduce computation and memory overhead on AR devices, we aim to compress the visual tokens in $\mathbf{T}_{\text{emb}}$, thereby reducing the key-value (KV) cache size required for later user queries and lowering the associated decoding cost.

The overall UTOPI pipeline, shown in Figure 3, consists of three stages: *Video Encoding*, *Adaptive Gaze-Aware Token Selector (AGTS)* and *Inter-Frame View Overlap Compressor (IVOC)*. After obtaining visual tokens $\mathbf{T}$ for all frames from the video encoding stage, we first apply the AGTS to each frame $\mathbf{I}^i$. The AGTS leverages real-time gaze information together with a learnable *Saliency Predictor*, to estimate token-level saliency and select foreground tokens related to the user’s attention, while discarding low-saliency background tokens. IVOC is then applied to these selected foreground tokens to further reduce temporal redundancy by using IMU-based pose estimation to identify overlapping regions between consecutive frames. Redundant overlap tokens are removed and replaced by a small set of clustered surrogate tokens, producing the final compressed visual tokens for downstream reasoning. We next describe the implementation details.

3.2 Adaptive Gaze-Aware Token Selector (AGTS)

According to Section 2, gaze location provides a strong prior for identifying semantically important tokens in egocentric visual systems. Modern AR devices are equipped with eye-tracking modules that record the 2D gaze coordinates (g_x^i, g_y^i) for each frame $\mathbf{I}^i$. However, simple gaze-based subsampling, such as cropping around the gaze location, is often insufficient to preserve strong LVU accuracy. To better exploit gaze information, we use it as a prior cue for token importance and employ a neural network to refine the importance scores for improved performance. Specifically, we apply a 2D Gaussian kernel centered at (g_x^i, g_y^i) on frame i :

$$G^i(w, h) = \exp\left(-\frac{(w - g_x^i)^2 + (h - g_y^i)^2}{2\sigma^2}\right) \quad (1)$$

where σ is a predefined standard deviation. Given this prior, we further introduce a saliency predictor, implemented as a lightweight MLP, which takes each visual token as input and produces an importance score $\alpha_{w,h}^i = \text{MLP}(\mathbf{T}_{w,h}^i)$. The final foreground tokens are then selected as:

$$\mathbf{T}_F^i = \{\mathbf{T}_{w,h}^i \mid \alpha_{w,h}^i \cdot G^i(w, h) > \rho\} \quad (2)$$

Here, ρ is a predefined threshold, and w, h denote the spatial location of token $\mathbf{T}_{w,h}^i$. The positional indices (i, w, h) of each foreground token are kept identical to the original, preserving spatiotemporal structure of foreground tokens.

To better align the user gaze distribution with the LVU task, we fine-tune the MLP using a calibration dataset constructed by randomly sampling 2k QA pairs from EgoEye while freezing

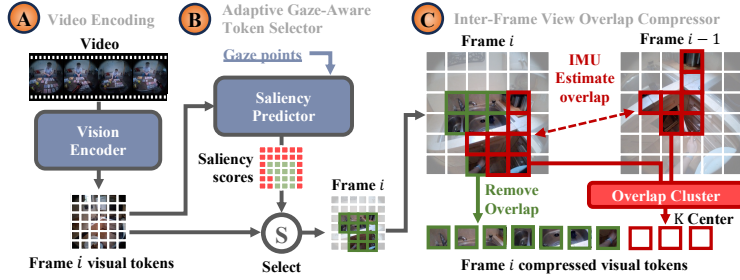


Figure 3: The overall workflow of UTOPI.

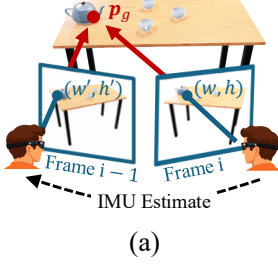


Figure 4: IMU based inter-frame coordinate projection.

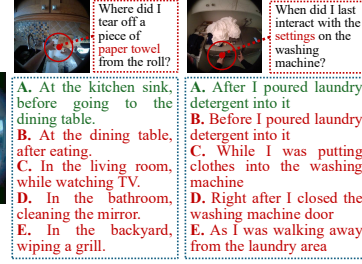


Figure 5: EgoEye QA example.

179 the remaining components of the VLM to reduce training cost. Specifically, during the forward
 180 pass, after obtaining $\mathbf{T}_F^i$ via Equation 2, we compute the final loss using next-token prediction
 181 $\mathcal{L}_{NTP} = -\log \frac{\exp(\mathbf{z}_y)}{\sum_{v=1}^V \exp(\mathbf{z}_v)}$, where $\mathbf{z} = F_{dec}(\mathbf{T}_F^i)$. $F_{dec}(\cdot)$ denotes the decoder function, $\mathbf{z} \in \mathbb{R}^V$
 182 represents the predicted logits over the token vocabulary, where V is the vocabulary size, and $\mathbf{z}_y$
 183 is the logit corresponding to the correct answer token. Since the language decoder is differentiable, we
 184 can compute $\partial \mathcal{L}_{NTP} / \partial \mathbf{T}_F^i$.

185 However, $\mathbf{T}_F^i$ is obtained from $\mathbf{T}_{w,h}^i$ via the hard selection defined in Equation 2 and is therefore not
 186 a differentiable function of the gate values α^i . As a result, $\partial \mathbf{T}_F^i / \partial \alpha^i$ is undefined. We thus adopt a
 187 straight-through estimator (STE) and approximate the gradient with respect to $\alpha_{w,h}^i$ as follows:

$$\frac{\partial \mathbf{T}_{w,h}^i}{\partial \alpha_{w,h}^i} \approx G^i(w, h) \sum_{d=1}^D \mathbf{T}_{w,h,d}^i, \forall \mathbf{T}_{w,h}^i \in \mathbf{T}_F^i \quad (3)$$

188 where $\mathbf{T}_{w,h}^i \in \mathbb{R}^D$ is a D -dimensional token embedding vector. Since α^i is produced by saliency
 189 predictor, we can backpropagate the gradients to the saliency predictor using Equation 3.

190 3.3 Inter-Frame View Overlap Compressor

191 Because egocentric videos are typically captured as a continuous first-person stream, they exhibit
 192 strong scene continuity over time, as illustrated in Figure 4. Content observed by the user persists
 193 across frames, while the corresponding tokens often shift spatially due to changes in head pose and
 194 viewpoint. For instance, Frame i and Frame $i-1$ in Figure 4 illustrate two consecutive frames from
 195 the AEA dataset at times $i-1$ and i , where the man and table are visible in both frames. We observe
 196 that corresponding visual tokens generally reappear at different spatial locations rather than remaining
 197 aligned at the same position. Although consecutive frames exhibit high temporal redundancy, directly
 198 comparing their RGB values often produces large differences. This is primarily caused by changes
 199 in the user’s head pose, which alter the camera’s orientation and position between frames. As a
 200 result, the same scene content may appear at different spatial locations in the image. Therefore, user
 201 pose information, which is naturally collected by modern AR devices, should be considered when
 202 assessing redundancy across frames.

203 To achieve this, we adopt a geometry-based reprojection strategy to estimate inter-frame overlaps.
 204 Specifically, for each foreground token $\mathbf{T}_{w,h}^i \in \mathbf{T}_F^i$ in the current frame $\mathbf{I}^i$ produced by AGTS
 205 module, its camera coordinate is denoted as (w, h) . As shown in Figure 4 (a), this coordinate can be
 206 mapped to a global coordinate $\mathbf{p}_g = \text{proj}_{c \rightarrow g}(w, h) \in \mathbb{R}^3$ using the camera intrinsics. Next, based
 207 on the camera pose change estimated from the IMU, we recover the camera pose of the previous
 208 frame $\mathbf{I}^{i-1}$ by reprojecting the global coordinate $\mathbf{p}_g$ back to that frame. This yields the corresponding
 209 camera coordinate $(w', h') = \text{proj}_{g \rightarrow c}(\mathbf{p}_g)$ in $\mathbf{I}^{i-1}$. We denote the mapping of camera coordinates
 210 between two frames as $f(\cdot, \cdot)$, such that $f(w, h) = (w', h')$. If (w', h') falls within the field of view
 211 of $\mathbf{I}^{i-1}$, we select the token $\mathbf{T}_{f(w,h)}^{i-1}$ in $\mathbf{I}^{i-1}$ located at this reprojected position. Specifically, we
 212 use the mapping $f(w, h) = (w', h')$ to index the corresponding token $\mathbf{T}_{f(w,h)}^{i-1}$ which represents the
 213 same visual region observed in the previous frame. Therefore, $\mathbf{T}_{f(w,h)}^{i-1}$ and $\mathbf{T}_{w,h}^i$ refer to the same
 214 physical region in the scene, observed from two consecutive frames. These corresponding regions
 215 together form the **overlapping area** between two frames, as illustrated in Figure 3 (c) by the red
 216 highlighted region. The detailed mapping formulas are provided in the Appendix. To reduce temporal

redundancy, we remove redundant tokens from overlapping visual regions and represent the frames using compressed **surrogate tokens**. The process is shown in Figure 3 (c). First, we identify the inter-frame overlap region $\mathbf{O}^i$ between frames $\mathbf{I}^i$ and $\mathbf{I}^{i-1}$ and collect the corresponding tokens in the current frame: $\mathbf{T}_O^i = \{\mathbf{T}_{w,h}^i | (w, h) \in \mathbf{O}^i, \mathbf{T}_{w,h}^i \in \mathbf{T}_F^i\}$. Notably, **IVOC is applied only to the foreground tokens selected by AGTS**, so $\mathbf{T}_O^i \subseteq \mathbf{T}_F^i$. We then map the spatial coordinates of these tokens to the previous frame $\mathbf{I}^{i-1}$ using the mapping $f(\cdot, \cdot)$. For each token $\mathbf{T}_{w,h}^i$, we compute the cosine similarity between it and its corresponding token at the mapped location in the previous frame:

$$s_{w,h}^i = \text{cosim}(\mathbf{T}_{f(w,h)}^{i-1}, \mathbf{T}_{w,h}^i), \quad (4)$$

We then identify tokens within $\mathbf{O}^i$ whose representations remain sufficiently similar to those in $\mathbf{I}^{i-1}$:

$$\mathbf{T}_S^i = \{\mathbf{T}_{w,h}^i \in \mathbf{T}_O^i \mid s_{w,h}^i > \theta\}, \quad (5)$$

where θ is the similarity threshold. Next, We find counterparts of $\mathbf{T}_S^i$ in the previous frame $\mathbf{I}^{i-1}$,

$$\mathbf{D}^{i-1} = \{\mathbf{T}_{f(w,h)}^{i-1} \mid s_{w,h}^i > \theta, (w, h) \in \mathbf{O}^i\}, \quad (6)$$

and cluster these tokens $\mathbf{D}^{i-1} \cup \mathbf{T}_S^i$ using k-means clustering to obtain K cluster centers $\{\mathbf{T}_C^k\}_{k=1}^K$, as illustrated by the *Overlap Cluster* module (highlighted in red) in Figure 3(c), which are used as surrogate tokens to replace $\mathbf{T}_S^i$. Finally, the surrogate tokens $\{\mathbf{T}_C^k\}_{k=1}^K$ and the retained foreground tokens $\mathbf{T}_F^i \setminus \mathbf{T}_S^i$ are used as the compressed visual tokens for the current frame. During subsequent VLM processing, for each surrogate token $\mathbf{T}_C^k$, we assign its temporal position to the current frame index i , and set its spatial position to the mean spatial coordinate of all tokens in the corresponding cluster. This coordinate is then mapped back to $\mathbf{I}^i$ via the inverse mapping $f^{-1}(\cdot, \cdot)$ and used as the final spatial location.

3.4 EgoEye Collection

While multiple benchmarks have been proposed for LVU, they primarily focus on generic video-based queries and do not fully capture the **human-centric** and **attention-guided** nature of real AR usage, where questions are closely tied to user attention patterns [66, 11, 26]. Motivated by this gap, and building on the background presented in Section 2., we introduce a new benchmark, termed *EgoEye*, for LVU that simulates realistic, attention-driven interactions in AR environments.

EgoEye is constructed from real-world egocentric video datasets with authentic gaze tracking, including Aria Everyday Activities (AEA) [39] and Nymeria [40]. In contrast to prior LVU benchmarks, our questions are explicitly **gaze-related**: we select question subjects by leveraging gaze traces to identify attended regions and objects, and then sample question targets with a strong bias toward items spatially close to the user’s fixation points over time. This design ensures that the queries reflect what the user actually looked at and interacted with during the experience. EgoEye includes video clips ranging from 50 seconds to 5 minutes, with an average length of 2 minutes and 31 seconds per clip. We extract more than 2k question subjects around users’ gaze locations, resulting in over 5k QA pairs. Figure 5 provides some QA examples for reference, with their question subjects being “Paper Tower” and “Washing Machine Knob”, respectively. Detailed statistics and more QA examples are provided in the Appendix.

4 Experiment

4.1 Evaluation Setups and Implementation Details

Video-QA Benchmarks To evaluate UTOPI, we use three Video Question–Answering (VQA) benchmarks: EgoSchema [44], EgoGazeVQA [52], and our proposed EgoEye. As UTOPI targets AR scenarios, all selected datasets are based on egocentric videos and multiple-choice questions. EgoSchema provides over 5k automatically constructed five-choice questions and offers a broad test of long-context understanding, reasoning, and retrieval. EgoGazeVQA supplies user gaze traces, with questions categorized into causal, spatial, and temporal reasoning. Both benchmarks are derived from the egocentric video dataset Ego4D [25]. Since UTOPI requires signals such as user gaze and IMU measurements, we obtain the original Ego4D metadata and align it to recover the necessary signals. For videos that does not have gaze signals available, we estimate the user gaze location using SUM [27], and further include an ablation study on the impact of gaze noise in Table 6.

Our EgoEye contains over 5k multiple-choice QA pairs (more detail in the Appendix) and includes user gaze and IMU metadata. In contrast to existing benchmarks, our questions are explicitly centered on human attention signals and realistic AR usage, producing queries that better reflect what users actually look at and care about, and enabling a more natural evaluation setting for long-context egocentric video understanding in AR environments.

Implementation Details For video input, we set the frame resolution to 532×532 and uniformly sample 300 frames per video for experiments on EgoEye and EgoSchema. For EgoGazeVQA, we sample 100 frames per video, prioritizing key frames that contain gaze points. To ensure a consistent token budget across all experiments for fair comparison, we adaptively adjust hyperparameters θ and ρ so that each video retains exactly the target ratio of visual tokens. We test the algorithms on the lightweight models including Qwen2.5VL-7B [10], Videollama3-7B [70], PLLaVA [63] and InternVL2.5-8B [15] and use four NVIDIA A6000 (48GB) GPUs for all accuracy evaluation. Finally, we set the number of cluster centers K to 2% of the total number of tokens in the cluster, i.e., $K = 0.02 \times |\mathbf{T}_S^i|$.

To evaluate UTOPI, we compare it against other methods for token pruning of LVU task, including PruneVid [28], DTD [64] and MindPalace [29]. We also evaluate *Random Drop* where the tokens are dropped randomly. For PruneVid [28], we disable its KV-cache compression module because it requires modifications to Qwen2.5VL and some implementation details are not publicly available. For MindPalace [29], we adopt Qwen2.5VL as the backbone LLM for generating final answers. When fine-tuning the saliency gating head, we randomly sample 2k QA pairs from EgoEye as the training set, using the remaining data for evaluation. During training, we interpret the gate values as probabilities for stochastic token dropping. We start with a low drop ratio and gradually increase it, while simultaneously annealing the temperature.

4.2 Accuracy Performance on LVU Tasks

In Table 1, we evaluate several VLM baselines and combinations of Qwen2.5VL with different compression methods on three benchmarks: EgoEye (EE), EgoSchema (ES), and EgoGazeVQA (EGV). All methods are evaluated under identical token budgets. The *Token Budget* in the table denotes the ratio of the number of visual tokens after compression to that before compression. A token budget marked with “*” indicates a value of 0%, meaning that only the question and answer options are provided to the model without any video input; this corresponds to a *blind test*. Some token budgets are marked with “†” to indicate these models do not apply token compression; instead, we downsample the video resolution to ensure that the total number of visual tokens matches ours.

Experimental results show that under the three token budgets (15%, 30%, 50%), UTOPI consistently outperforms all other efficiency baselines across all benchmarks. Specifically, when applied to Qwen2.5VL, UTOPI retains only 15% of the visual tokens while still improving accuracy by an average of 2.7% compared to using the full set of visual tokens. Compared with other compression methods, UTOPI achieves at least 2.5% higher accuracy on average. In addition, relative to naive uniform-resolution downsampling of Qwen2.5VL, UTOPI improves the average accuracy by 8.6%. Finally, compared with the blind test setting, UTOPI uses only 15% of the visual tokens yet achieves an average accuracy gain of 15%, further demonstrating its effectiveness.

4.3 UTOPI System Performance

Latency and Memory To evaluate the system performance of UTOPI, we implement the system on both a desktop GPU (RTX A6000) and an edge GPU (NVIDIA Jetson Orin NX), and report latency and peak memory usage. These metrics capture the responsiveness and resource efficiency of the

Table 1: Acc. Evaluation of UTOPI.

Method	Token Budget	EE	ES	EGV
QWen2.5VL-7B	100%	58.4%	60.1%	47.8%
QWen2.5VL-7B	0%*	36.6%	49.4%	43.4%
<i>QWen2.5VL-7B</i>				
w/ UTOPI (ours)	15%	60.6%	64.8%	49.0%
w/ DTD [64]	15%	57.6%	64.5%	47.9%
w/ PruneVid [28]	15%	51.3%	63.0%	48.8%
w/ Random Drop	15%	56.8%	59.3%	46.1%
QWen2.5VL-7B	15%†	50.4%	53.8%	44.2%
MindPalace [29]	15%†	33.2%	45.2%	37.9%
PLLaVA [63]	15%†	34.1%	29.6%	31.5%
Videollama3-7B [70]	15%†	46.4%	49.7%	42.5%
InternVL2.5-8B [15]	15%†	52.4%	51.5%	43.0%
<i>QWen2.5VL-7B</i>				
w/ UTOPI (ours)	50%	57.1%	61.9%	48.5%
w/ DTD	50%	57.0%	60.2%	47.2%
w/ PruneVid	50%	52.4%	60.4%	47.8%
w/ Random Drop	50%	56.4%	59.6%	46.9%
w/ UTOPI (ours)	30%	57.4%	62.6%	48.7%
w/ DTD	30%	56.6%	62.1%	48.2%
w/ PruneVid	30%	53.3%	61.9%	48.1%
w/ Random Drop	30%	56.1%	60.3%	45.8%

Table 2: Latency and Peak Memory Evaluation. "w/ ours" indicates that UTOPI is implemented over the corresponding model.

Method	platform	Token budget	Peak Memory Cost (w/o Model weights)			UTOPI Overhead		TTFT (speed up $\times$)	TPS (speed up $\times$)	EE Acc.
			VE	LD	Overall (save%)	MFLOPs	Latency			
QWen 7B	GPU	100%	6.6G	19.8G	19.8G	–	–	19.3s	7.5	58.4%
QWen 7B w/ ours	GPU	15%	6.6G	4.1G	6.6G (67%)	1636	673ms	8.7s (2.2 $\times$)	26.6 (3.5 $\times$)	60.6%
QWen 7B (4bit)	Jetson Orin	100%	3.2G	7.1G	OOM 7.1G	–	–	OOM 173s	OOM 0.43	57.3%
QWen 7B (4bit) w/ ours	Jetson Orin	15%	3.2G	1.8G	3.2G (55%)	545	980ms	88s (2.0 $\times$)	1.3 (3.0 $\times$)	57.5%
QWen 3B (4bit)	Jetson Orin	100%	2.9G	6.4G	6.4G	–	–	122s	0.73	52.0%
QWen 3B (4bit) w/ ours	Jetson Orin	25%	2.9G	2.2G	2.9G (55%)	545	980ms	65s (1.9 $\times$)	2.0 (2.7 $\times$)	52.1%
SmolVLM 2.2B	Jetson Orin	100%	2.0G	2.9G	2.9G	–	–	21s	3.2	46.3%
SmolVLM 2.2B w/ ours	Jetson Orin	60%	2.0G	1.8G	2.0G (32%)	360	440ms	13s (1.6 $\times$)	4.8 (1.5 $\times$)	45.9%

system, which are critical for real-time AR workloads deployed on resource-constrained edge devices. NVIDIA Jetson Orin NX has been widely used in prior work [50, 4, 3] to simulate AR/VR devices. This platform provides 16GB of unified memory. Since the VLMs evaluated in Table 1 cannot fully fit within the memory capacity of the edge GPU, we adopt 4-bit quantized Qwen2.5VL-7B [61] and Qwen2.5VL-3B [61] as baselines, using publicly available checkpoints from Hugging Face. We also evaluate UTOPI on SmolVLM [45].

Table 2 reports the *Peak Memory Cost*, defined as the maximum memory usage during inference. We exclude the static memory occupied by model weights and report only the dynamic components, such as layer activations and the KV cache. **VE** denotes the peak memory cost of the Vision Encoder, while **LD** denotes that of the Language Decoder. **Overall** corresponds to the maximum of the two (highlighted in yellow). We observe that without compression, LD becomes the primary bottleneck due to the large KV cache, with memory usage on average 2.2 $\times$ higher than that of the VE. With UTOPI, while maintaining comparable accuracy, the peak memory consumption is reduced by an average of 52.2%. In particular, Qwen2.5VL-7B (4-bit) runs out of memory without the introduction of UTOPI (the red entry in the corresponding row represents a reasonable estimate).

Additionally, we report latency results for UTOPI in terms of *Time To First Token* (TTFT), which measures the time from input to the first generated token, and throughput, i.e., Tokens Per Second (TPS), in Table 2. Similar metrics have been used in prior work to evaluate the system performance of large models [24, 68, 5]. The results show that both response latency and decoding throughput improve substantially, achieving average gains of 1.92 $\times$ and 2.67 $\times$, respectively. Overall, these results demonstrate that UTOPI significantly reduces memory and latency bottlenecks while preserving accuracy, making large VLMs more practical for deployment in AR environments.

4.4 Ablation Studies

Performance Under Different Budget To evaluate the trade-off between accuracy and token budget, we apply UTOPI to PLLaVA [63] and Qwen2.5VL-7B and compare results under different token budgets on two benchmarks (Table 3). A token budget of 100% denotes the original model. The best accuracy is highlighted in bold, and the smallest token budget that matches or exceeds the baseline accuracy is underlined. Results show that compressing visual tokens with UTOPI often improves accuracy on EgoSchema, likely because removing redundant tokens reduces interference. With Qwen2.5VL, our method compresses up to 95% of visual tokens without accuracy loss. On PLLaVA with EgoSchema, UTOPI achieves up to 85% token reduction while maintaining performance. On EgoEye, accuracy drops by less than 2% at a 15% token budget. Overall, UTOPI achieves a strong balance between efficiency and accuracy.

Table 3: Budget Analysis.

Baseline	Token Budget	EE	ES
PLLaVA	100%	34.8%	37.8%
w/ UTOPI	50%	33.7%	39.8%
w/ UTOPI	30%	34.2%	40.1%
w/ UTOPI	15%	33.5%	39.3%
w/ UTOPI	5%	31.8%	32.1%
QWen2.5VL-7B	100%	58.4%	60.1%
w/ UTOPI	50%	57.1%	61.9%
w/ UTOPI	30%	57.4%	62.6%
w/ UTOPI	15%	60.6%	64.8%
w/ UTOPI	5%	53.4%	61.4%
w/ UTOPI	3%	50.5%	59.4%

Impact of AGTS and IVOC To validate the effectiveness of the AGTS and IVOC modules, we conduct ablation studies, as shown in Table 4. We evaluate Qwen2.5VL-7B with UTOPI under different configurations on EgoSchema. The results show that enabling either AGTS or IVOC improves accuracy to around 64%. When both are enabled, accuracy reaches 64.8% while achieving the lowest token budget. Since AGTS and IVOC address spatial and temporal redundancies, respectively,

Table 4: Evaluation of different AGTS / IVOC configurations

AGTS	IVOC	Token Budget	ES
✓	✓	15%	64.8%
✓	×	50%	64.1%
×	✓	30%	64.4%
×	×	100%	60.1%

Table 5: Performance comparison on the Low-Motion subset “SLOW” and Motion-Intensive subset “FAST”.

Method	Token Budget	ES		
		FAST	SLOW	AVG
QWen w/ ours	30%	62.0%	63.2%	62.6%
QWen w/ ours	15%	63.6%	66.0%	64.8%
QWen w/ ours	5%	59.2%	63.6%	61.4%

Table 6: Impact of gaze noise.

Gaze Noise	EE	EGV
0 px	60.6%	49.0%
75 px	60.3%	48.7%
150 px	60.4%	47.7%
225 px	58.9%	47.4%
300 px	57.7%	47.1%

they complement each other and jointly improve compression efficiency without harming accuracy, enabling both lower token budgets and better overall performance.

Impact of Motion-Intensive Video While most videos in the test set contain egocentric footage with typical human motion, we assess IVOC’s robustness under unusual conditions by comparing Qwen2.5VL-7B with UTOPI on *Motion-Intensive* and *Low-Motion* videos (see Table 5). To construct the evaluation split, we first encode each EgoSchema video with the QWen2.5VL vision encoder and obtain a token map for every frame, denoted $\mathbf{T}^i$. We then average all token embeddings within each frame to get a single frame-level representation, $\mathbf{r}_i = \text{mean}(\mathbf{T}^i)$. Next, we compute the cosine similarity between consecutive frame representations and average over the video to obtain the mean inter-frame similarity, namely $\alpha = \frac{1}{N} \sum_{i=1}^N \cos(\mathbf{r}_i, \mathbf{r}_{i+1})$. After that, we sort all EgoSchema videos by α and select the top 50% (larger α) as *Low-Motion* videos (denoted as SLOW); the remaining 50% are *Motion-Intensive* videos (denoted as FAST).

As shown in Table 5, under the same token budget, SLOW videos generally achieve higher accuracy than FAST ones. However, the gap is small, averaging 2.6%, indicating that UTOPI remains robust across videos with different levels of temporal redundancy.

Impact on Gaze Noise Finally, we evaluate the impact of gaze tracking noise on UTOPI. Specifically, we inject perturbations of varying magnitudes into the gaze signals in EgoEye and EgoGazeVQA, ranging from 75 pixels to 300 pixels. For instance, at a noise level of 75 pixels, the perturbed gaze point is uniformly sampled within a circle centered at the ground-truth gaze point with a radius of 75 pixels. Table 6 reports the results, where the baseline is Qwen2.5-VL-7B with a token budget of 15%. All other settings follow Table 1. Even under large gaze noise, the accuracy drop remains within 3%. These results show the robustness of our method.

Additionally, when gaze tracking is unavailable, UTOPI can still operate by estimating gaze locations with a gaze predictor, such as SUM [27], as discussed in Section 4.1. To evaluate this setting, we replace the ground-truth gaze locations in EgoEye with SUM-predicted gaze locations and repeat the accuracy evaluation. Under a 15% token budget, ground-truth gaze achieves 60.6% accuracy on EE, while predicted gaze achieves 59.4%, with a gap of less than 1.2%. This small difference shows that UTOPI remains effective even when direct gaze tracking is unavailable.

5 Conclusion and Limitation

We propose UTOPI, a plug-and-play user behavior guided token pre compress module in AR environments. UTOPI improves efficiency by using gaze-informed attention to discard spatially redundant visual tokens. Experiments show that UTOPI compresses up to 95% of visual tokens on Qwen2.5VL and 85% on PLLaVA while maintaining baseline accuracy. It achieves the highest accuracy (64.8%) under the strongest compression and remains robust across motion levels, with only a 2.6% gap. One limitation is that the current evaluation of UTOPI focuses on egocentric LVU benchmarks and AR simulation platforms; future work will further validate UTOPI on real-time AR hardware and more diverse deployment settings.

396 A Technical appendices and supplementary material

397 A.1 EgoEye dataset

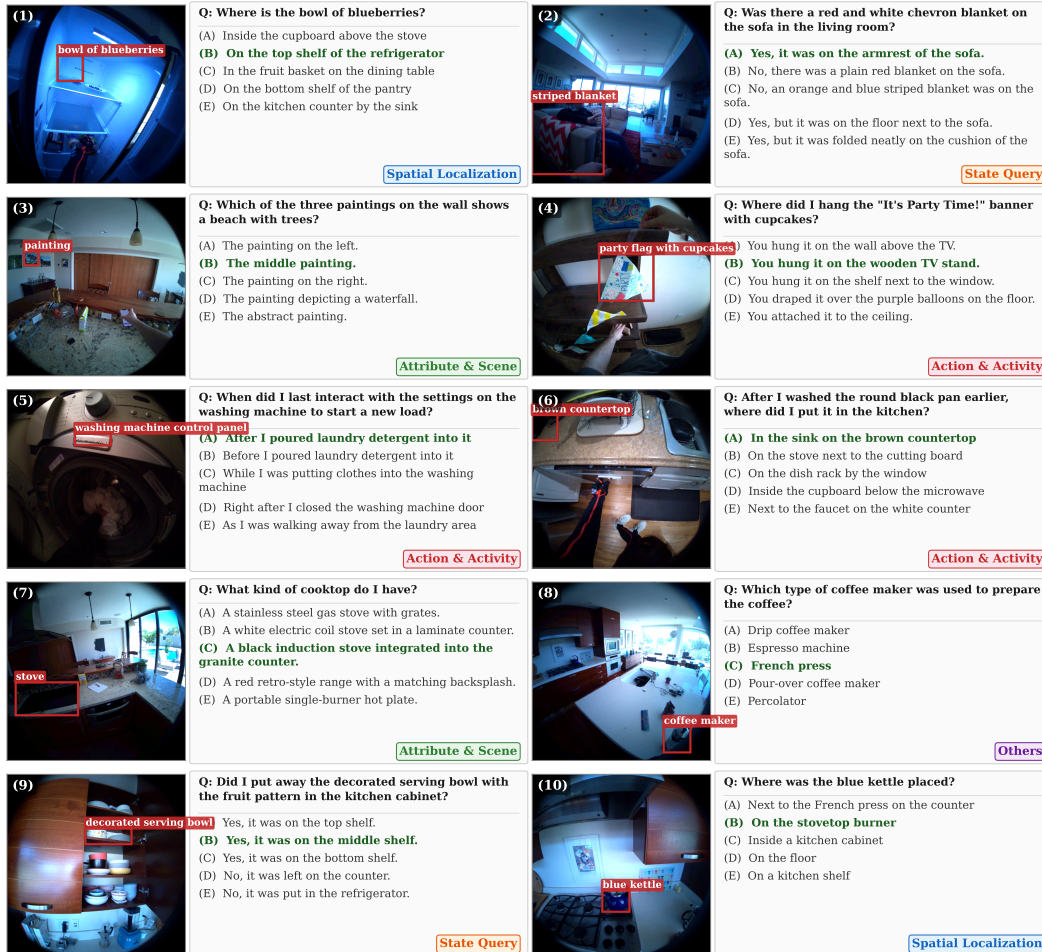


Figure 6: Additional question-option examples in EgoEye and their corresponding selected question subjects.

398 The videos in EgoEye are drawn from two public datasets, Aria Everyday Activities (AEA) and
 399 Nymeria. Figure 7 shows the distribution of video durations in EgoEye. Based on these videos,
 400 we generated approximately 5K question-answer (QA) pairs. To construct the QA pairs, we first
 401 sample keyframes from each video and feed them, together with the user gaze coordinates at the
 402 corresponding timestamps, into a VLM. The VLM is then asked to identify an object around the
 403 gaze point as the question subject. Next, we temporally downsample the video, using a higher
 404 sampling rate for frames closer to the keyframe, and require the VLM to generate a question about the
 405 question subject, along with the corresponding options and answer. By summarizing the generated
 406 questions, we roughly group them into five high-level categories: (1) Action & Activate, which
 407 focuses on dynamic questions such as action recognition, event ordering, and object tracking; (2)
 408 State Query, which includes yes-or-no style questions about the state of an object or scene, such as
 409 whether something is on or off, or present or absent; (3) Attribute & Scene, which covers appearance
 410 descriptions such as color and material, as well as scene details; (4) Spacial Localization, which
 411 focuses on spatial positioning; and (5) Others, which includes remaining types of questions, such
 412 as counting. Fig. 6 provides several examples of Question-Options from each category, together
 413 with their corresponding keyframes and question subjects (highlighted by red boxes). Fig. 8 shows
 414 the proportion of each category, as well as the accuracy of QWen 2.5 VL 7B with various token
 415 compression method at a 15% token budget across different categories.

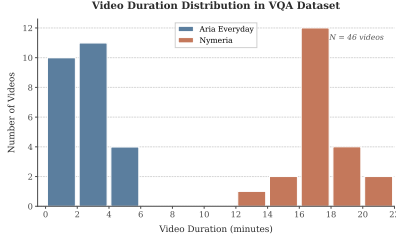


Figure 7: Video duration distribution in EgoEye.

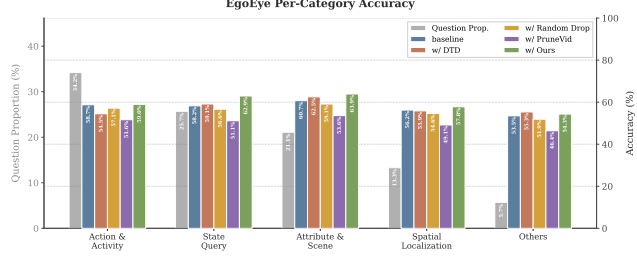


Figure 8: EgoEye Category distribution and per-category Accuracy.

A.2 IVOC re-projection

For a token $\mathbf{T}_{w,h}^i$ in frame $\mathbf{I}^i$, let (u, v) denote the center pixel coordinate of its corresponding image patch. Here, W_{img} and H_{img} are the image width and height, while W and H are the width and height of the token grid. The camera intrinsic matrix is denoted by $\mathbf{K}$, where $\mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$, with f_x, f_y being the focal lengths and (c_x, c_y) being the principal point. $(\mathbf{R}_i, \mathbf{t}_i)$ denotes the 6-DoF camera pose of frame $\mathbf{I}^i$, obtained from the AR device’s pose tracking system, where $\mathbf{R}_i$ and $\mathbf{t}_i$ represent the camera rotation and translation in the global coordinate system, respectively.

The corresponding global 3D coordinate is computed as

$$\mathbf{p}_g = \mathbf{R}_i \left(d \mathbf{K}^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \right) + \mathbf{t}_i. \quad (7)$$

d denote the depth value at (u, v) . For a few videos missing IMU or depth signals, we get camera motion and depth using MegaSaM [35] and use it as the sensor input from the AR/VR device. We then reproject $\mathbf{p}_g$ into frame $\mathbf{I}^{i-1}$:

$$\begin{bmatrix} \tilde{u}' \\ \tilde{v}' \\ \tilde{z}' \end{bmatrix} = \mathbf{K} \mathbf{R}_{i-1}^\top (\mathbf{p}_g - \mathbf{t}_{i-1}), \quad u' = \frac{\tilde{u}'}{\tilde{z}'}, \quad v' = \frac{\tilde{v}'}{\tilde{z}'}. \quad (8)$$

After discretization, the mapped token index is

$$f(w, h) = (w', h'), \quad w' = \left\lfloor u' \frac{W}{W_{\text{img}}} \right\rfloor + 1, \quad h' = \left\lfloor v' \frac{H}{H_{\text{img}}} \right\rfloor + 1. \quad (9)$$

Accordingly, the valid overlap region is

$$\mathbf{O}^i = \{(w, h) \mid \tilde{z}' > 0, 0 \leq u' < W_{\text{img}}, 0 \leq v' < H_{\text{img}}\}. \quad (10)$$

The inverse mapping $f^{-1}(\cdot, \cdot)$ is obtained in the same manner by swapping i and $i - 1$.

A.3 Visualization

We provide several visualization clips in Fig. 9 for reference. (a) shows the IVOC token drop, where the black mask indicates dropped tokens. (b) shows the AGTS saliency heat map of corresponding frame.

A.4 Impact of Object-Gaze Distance

To evaluate the robustness of the AGTS, we group questions in the EgoEye benchmark by the distance between the question subject and the gaze point. We then compare the accuracy of Qwen2.5VL-7B with different compression methods across these distance groups (Figure 10). Note that EgoEye frames have a resolution of 1408×1408 . As the distance increases, all models exhibit lower accuracy,

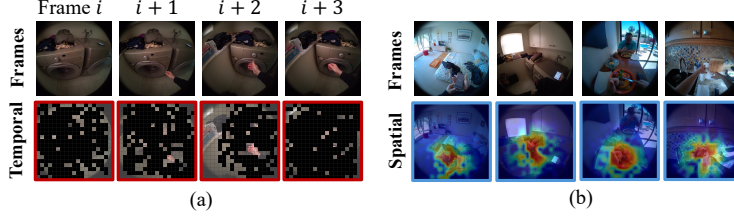


Figure 9: Visualization of AGTS and IVOC.

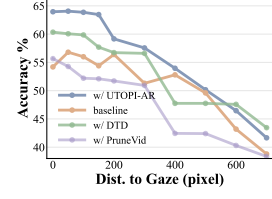


Figure 10: Accuracy vs. question-gaze distance.

likely because objects far from the gaze point tend to appear near the frame boundary, where visual cues are less stable and questions become more challenging. When the question subject is close to the gaze point, UTOPI achieves about 5% higher accuracy, demonstrating the benefit of AGTS in leveraging user attention to preserve informative regions. Even when the subject is far from the gaze point, UTOPI maintains comparable accuracy to other methods, indicating strong robustness.

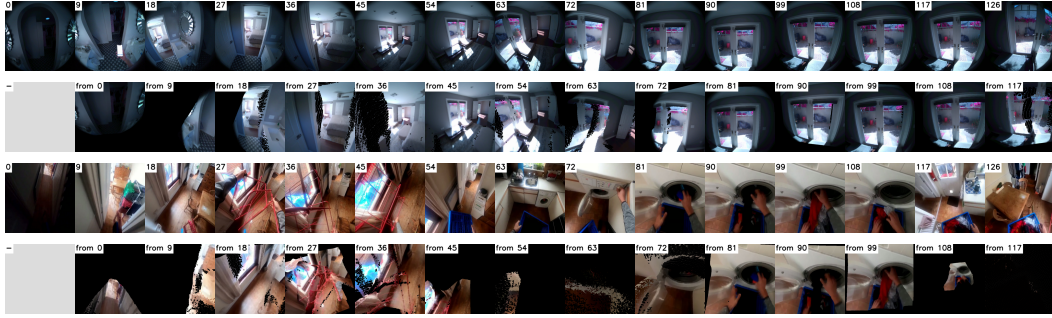


Figure 11: The visualization of re-projection. The odd rows show the original sampled frames, while the even rows show the previous sampled frames re-projected onto the current frames. To better illustrate the re-mapping quality, we present pixel-level re-projection here. As can be seen, the frames obtained by mapping from the previous frame closely match the original frames. The differences between the upper and lower frames mainly come from two factors: (1) the black regions indicate locations that are not visible in the previous frame; (2) dynamic contents, such as Frame 27 in the second video and the frame labeled “from 18”.

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